Parth Nitesh Chaudhari

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Education

International Institute of Information Technology, Hyderabad — Hyderabad, India
B.Tech in Computer Science and Engineering — *July 2024 – May 2028*

Experience

Undergraduate Research Assistant — April 2026 – Present
Data Science and Analytics Centre, IIIT-H (Prof. Kamalakar Karlapalem) — *Python, pandas, scikit-learn*

- Studied how a defence's shape affects chance creation across **17 pro matches** in two leagues (SkillCorner A-League, DFL Bundesliga): built **8,087** phases of play with 339 shots and 37 goals, and wrote rule-based phase labels for the league that had none.
- Measured 9 shape features every 0.5 s and tested each with tertile splits, logistic models adjusted for ball position and league, and **1,000-resample** bootstraps over games, then replicated every headline finding in the second league.
- Key result: with few defenders behind the ball, **51% vs 27%** of counter-attacks ended in a shot (A-League) and xG per attack was **7.4**× higher (Bundesliga); a shape-only model reached AUC 0.47–0.63 on unseen games, so shape shifts odds rather than deciding single attacks.
- Labelled **24,374** passing options from player geometry and validated the rule on 5,803 held-out passes (**86% vs 62%** completion, clear lane vs 2+ defenders); showed a defender count beats a yes/no flag (AUC 0.61 vs 0.53).
- Diagnosed that a video pipeline's danger score (36,750 frames) followed the camera rather than the game, and moved the study to pitch-coordinate tracking data.

Projects

LLM Confidence Under Pressure | *Python, PyTorch, Hugging Face, W&B* — September 2026 – October 2026

- Designed a controlled experiment: two open LLMs, three feedback conditions (criticism, praise, neutral control), six turns, **450 claims** with known truth, each true claim paired with a false twin.
- Under criticism Llama-3.1-8B kept **14.7%** of correct MATH answers vs **68.6%** in the control, and its true/false discrimination fell from 0.46 to **0.11**; Qwen2.5-7B's own working still reached the right answer in **73%** of its reversals. Bootstrap CIs with fixed seeds throughout.

Global Earthquakes, 2015–2025 | *D3.js, JavaScript* — March 2026

- 3-person team: interactive D3 views of **80,829** USGS earthquake records, including a hexbin density map with tectonic plate overlay, a depth vs magnitude density plot and a map of monitoring coverage and data quality.

Cosmic 3D: IoT Printer Dashboard | *TypeScript, React, MQTT, MongoDB* — January 2026 – May 2026

- 5-person client project (DASS course); owned the MQTT layer that streams live printer telemetry from ESP32 firmware to a React dashboard.

Event Management Platform | *React, Express, MongoDB, Socket.IO* — February 2026

- Built solo and deployed a **7,800-line** full-stack platform: **52 REST endpoints**, JWT role-based portals, real-time Socket.IO chat, QR e-tickets by email and a custom form builder.

CLI-Docs: Distributed Document Store | *C, pthreads, TCP sockets* — October 2025 – November 2025

- **4,700-line** C system (2-person team): name server, multi-threaded storage servers with per-sentence write locks, O(1) hash lookups, persisted metadata and undo.

Institute Management Android App | *Kotlin, Jetpack Compose, Material 3* — April 2026

- 2-person team; wrote the **5,500-line** Kotlin codebase (26 files): role-aware dashboards for students and faculty, OTP sign-in flow, 24-hour timetable with alerts and per-course attendance marking.

Technical Skills

Languages: Python, SQL, JavaScript, TypeScript, C
Data: pandas, NumPy, scikit-learn, Matplotlib, D3.js, OpenCV, Jupyter
Methods: logistic regression, odds ratios, bootstrap confidence intervals, cross-validation, experiment design
Tools: Git, Linux, Slurm, Weights & Biases, PyTorch, Hugging Face